A Lightweight Dual-Branch Network for Building Change Detection in Remote Sensing Images Integrating Cross-Scale Coupling and Boundary Constraint

Yanshuai Dai, Li Shen, *Member, IEEE*, Yong Wang, Shichuan Liu, and Zhilin Li

***Abstract*— Capturing spatial details and contextual semantics is crucial for building change detection (BCD) in remote sensing images. However, achieving both aspects within traditional single-branch feature extraction networks faces challenges in computation costs and model sizes. To tackle these challenges, this article introduces a lightweight neural model tailored for BCD in remote sensing images. Its primary contribution lies in the design of a lightweight dual branch for efficient feature extraction, a cross-scale coupling module (CSCM) for effective multiscale feature enhancement, and a boundary constraint module (BCM) for edge details compensation. Specifically, the lightweight dual branch can efficiently extract spatial details and contextual semantics in two independent branches, thereby generating changed detail-semantics feature maps. The CSCM further enriches the semantic and scale representation ability of changed feature maps, by adapting to the multiscale characteristics of changed buildings. In addition, the BCM can improve the model's sensitivity to the boundaries of changed buildings, mitigating the loss of edge information caused by convolution and pooling operations. As a result, the fine-grained and high-level semantic feature maps for BCD are obtained. We comprehensively evaluate the effectiveness of our proposed method against numerous state-of-the-art lightweight and non lightweight change detection models on the LEVIR and WHU datasets. The results demonstrate that our method not only achieves remarkable accuracy but also stands out in efficiency.**

***Index Terms*— Boundary constraint module (BCM), building change detection (BCD), contextual semantics, cross-scale coupling module (CSCM), dual branch, lightweight network, spatial details.**

I. Introduction

BUILDING change detection (BCD) in remote sensing images aims to identify changes in buildings within the same region by comparing images captured at different times. It holds a crucial role in environmental monitoring, urban planning, and damage surveys [1], [2], [3].

Recently, with its powerful feature expression capacities, deep learning has achieved remarkable performance in BCD from very high-resolution remote sensing images [4], [5], [6]. In general, the mainstream approach for deep learning-based BCD involves treating it as a binary semantic segmentation task to achieve the identification of changed buildings [7], [8], [9]. For example, Liu et al. [10] proposed a densely attentive refinement network employing the U-shape encoder–decoder architecture for multiscale changed buildings detection. Likewise, Ding et al. [11] presented a deep encoder–decoder architecture that integrates a spatial attention mechanism-guided cross-layer addition and skip-connection module, alongside an atrous spatial pyramid pooling module for BCD. In addition, by leveraging the transformer architecture, Chen et al. [12] raised an edge-guided block and a feature differential enhancement module to generate highly discriminative changed feature maps for achieving fine-grained BCD. Furthermore, Xue et al. [13] proposed a multifrequency channel attention module and a joint multifrequency difference feature enhancement guiding block to improve the ability of channel correlation and the boundary quality of changed objects for change detection, respectively.

It can be seen that although auxiliary modules, such as multiscale, attention mechanism, and boundary guidance, have improved performance to some extent, their success is primarily attributed to the adoption of deep and wide network architectures. In such network architectures, spatial details and contextual semantics can be captured simultaneously, thereby generating fine-grained and high-level semantic changed feature maps. Nevertheless, capturing spatial details requires convolutional layers with wide channels, whereas understanding contextual semantics demands a network with sufficient depth. Therefore, achieving these goals in the single-branch network architectures poses challenges in terms of computational cost and model size [14], [15]. It is unavoidable to limit their widespread popularization for practical applications, particularly for deployment on unmanned aerial vehicles (UAVs) or satellites with limited energy and computational capability [16], [17]. As a result, there is an urgent demand to develop lightweight network architectures with less computation cost and smaller model sizes to facilitate BCD

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research.

Nowadays, various well-known and efficient lightweight networks, such as MobileNets [18], ShuffleNets [19], Xception [20], and EfficientNet [21], have gradually been adopted into change detection tasks. For example, to decrease the network parameters, Yang et al. [22] raised a lightweight BCD network utilizing MobileNetV2 to extract the bitemporal features and introduced the multilevel feature fusion module to recover the multiscale changed buildings maps. Also, to improve the computation efficiency, Wang et al. [23] used the EfficientNet as the backbone and proposed a feature interleaved fusion module for interleaving the shared features and the difference variance features for BCD. In addition, Fang et al. [24] proposed a MetaChange framework, which integrates interactive alignment and feature fusion for change detection research. Furthermore, Jiang et al. [25] proposed an efficient visual change transformer for change detection, where each pixel of the feature maps is treated as a graph node to create a graph neural network; then, reliable tokens are strengthened by employing self/cross attention schemes. While the aforementioned networks have undoubtedly advanced lightweight change detection, they predominantly utilize techniques such as depthwise separable convolution, pointwise group convolution, inverse residual blocks, or lightweight transformers as substitutes for traditional heavy convolutions. Essentially, they still retain deep and wide network architectures without fundamentally addressing the challenges of computation costs and model sizes [26], [27], [28].

Afterward, some scholars have advocated for a dual-branch architecture as a means to achieve lightweight networks for semantic segmentation [14], [15], [26], [27], [28]. In this architecture, spatial details and contextual semantics are encoded separately in two independent branches to overcome the shortcomings of deep and wide network architectures. One branch utilizes a wide-channel, small receptive field, and shallow convolutional structure to quickly extract rich spatial information, whereas the other branch employs a narrow-channel, large receptive field, and sufficient-depth convolutional structure to efficiently capture high-level semantic information. By utilizing the dual-branch architecture, the challenges related to high computation costs and large model sizes can be effectively mitigated, while the simultaneous capture of spatial information and contextual semantics becomes feasible.

Consequently, inspired by this idea, we propose a lightweight dual-branch network for BCD. It comprises a lightweight dual branch for efficient feature extraction, a cross-scale coupling module (CSCM) for effective multiscale feature enhancement, and a boundary constraint module (BCM) for edge details compensation. Specifically, in the lightweight dual branch, one branch, referred to as the "detail branch," adopts three simple convolutional stages with wide channels and small receptive fields to maintain the high-resolution spatial feature maps of changed buildings. The other branch, referred to as the "semantics branch," employs a fast downsampling strategy based on depthwise convolution, residual connection, and global average pooling to efficiently capture high-level semantic feature maps. Concurrently, the detail-semantics aggregation module (DSAM) addresses the gaps in spatial resolution and semantic level between the two branches, thereby generating the changed detail-semantics feature maps. Furthermore, aiming at the multiscale characteristics of changed buildings resulting from partial expansion, reduction, construction, and demolition, we propose a CSCM to further enrich the semantic and scale representation ability of changed feature maps. In addition, considering the loss of edge information caused by convolution and pooling operations, as well as the complexity of the edge due to the accompanying shadows and occlusion from other land covers, we introduce a BCM to improve the model's sensitivity to the boundaries of changed buildings. As a result, the fine-grained and high-level semantic feature maps for BCD are obtained.

The main contributions can be summarized as follows.

1) A novel lightweight dual-branch BCD network architecture for efficiently capturing spatial details and contextual semantics is proposed. This architecture effectively addresses the challenges related to high computation costs and large model sizes while enabling the generation of fine-grained and high-level semantic changed feature maps.
2) Built on the dual-branch architecture, a CSCM for effective multiscale feature enhancement is embedded to enrich the semantic and scale representation ability of changed feature maps.
3) Moreover, a BCM for edge details compensation is employed to further improve the model's sensitivity to the boundaries of changed buildings.

The rest of this article is organized as follows. Section II describes the proposed method. Section III shows the experimental results and analysis. Finally, the discussion and conclusion are drawn in Sections IV and V, respectively.

II. Proposed Method

As shown in Fig. 1, our proposed lightweight BCD network includes a dual-branch architecture, a CSCM, and a BCM.

A. Dual-Branch BCD Architecture

Our dual-branch network can be considered a derivative form of Siamese networks, comprising a detail branch, a semantics branch, and a DSAM. In traditional Siamese networks, bitemporal images are input separately into a backbone network to generate bitemporal feature maps, which are then compared to capture change information. Similarly, in this article, the backbone network comprises a detail branch and a semantics branch. The detail branch extracts bitemporal spatial details, while the semantics branch captures bitemporal multilevel contextual semantics. The differences between bitemporal spatial details and bitemporal multilevel contextual semantics reveal changed spatial details and changed multilevel contextual semantics, respectively. Next, the DSAM is introduced to address the gaps in spatial resolution and semantic level between the two branches, thereby obtaining the changed detail-semantics feature maps.

Specifically, the detail branch consists of only three stages, each comprising two or three convolutional layers, followed

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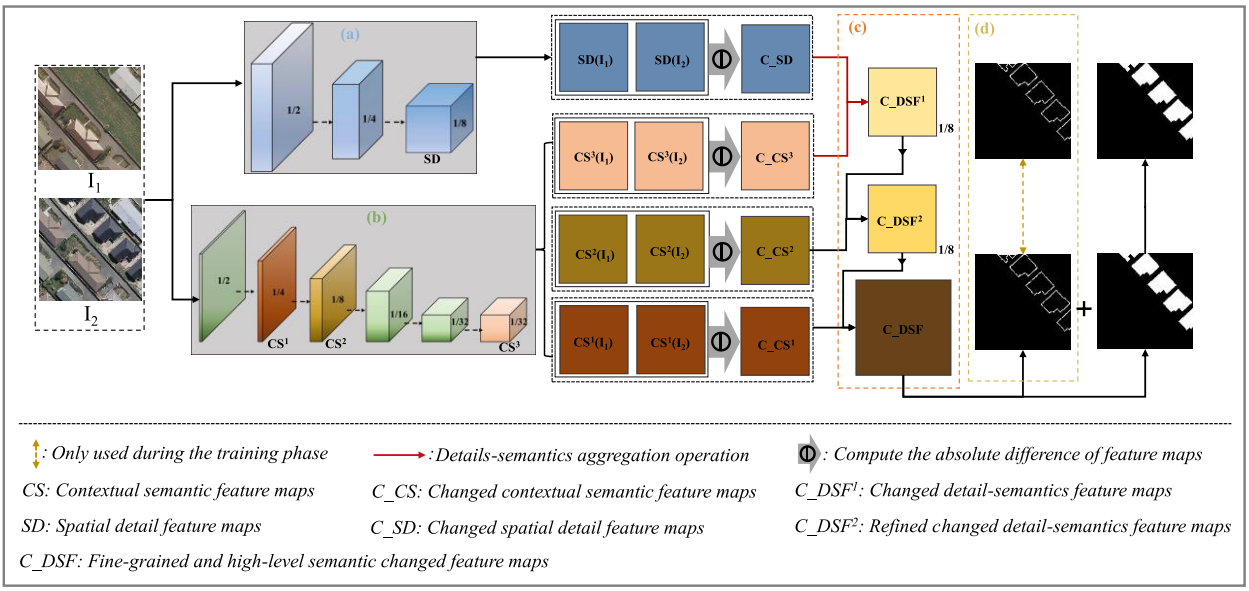


Fig. 1. Overall framework of the proposed method. (a) Detail branch. (b) Semantics branch. (c) Cross-scale coupling. (d) Boundary constraint.

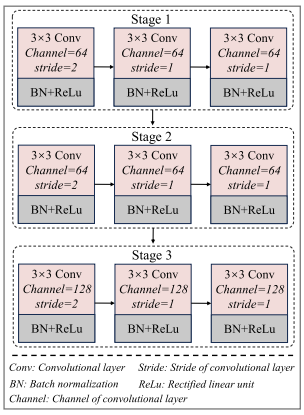


Fig. 2. Components of the detail branch.

by batch normalization and an activation function, as detailed in Fig. 2. Therefore, for a pair of images $I1$ and $I2$ with $C \times H \times W$, where C, H, and W represent the number of channels, height, and width in original images, respectively, we can also obtain the spatial details $SD(I1)$ and $SD(I2)$ with $128 \times H/8 \times W/8$, respectively. Finally, the changed spatial details C_SD can be acquired by calculating the absolute value based on the corresponding channels of $SD(I1)$ and $SD(I2)$. It can be seen that C_SD is generated through a simple convolutional structure with 0.5 M parameters.

Compared to the detail branch, the semantics branch is more complex, comprising a stem block, six gather-and-expansion blocks, and a context embedding block, as illustrated in Fig. 3. Among them, the stem block employs two different downsampling manners to improve feature expression ability; and the gather-and-expansion block utilizes the depthwise convolution to efficiently extract and fuse the contextual semantics between in the high-dimensional space and the low-dimensional space, and the context embedding block employs the global average pooling and residual connection to inset the global contextual information. For an image $I1$ with $C \times H \times W$, we can obtain the multilevel contextual semantics of CS^1, CS^2, and CS^3 with $128 \times H/4 \times W/4$, $128 \times H/8 \times W/8$, and $128 \times H/32 \times W/32$, respectively. Therefore, for a pair of images $I1$ and $I2$, the changed multilevel contextual semantics C_CS^1, C_CS^2, and C_CS^3 can be acquired by calculating the absolute value based on the corresponding channels of $CS^1(I1)$, $CS^2(I1)$, $CS^3(I1)$, $CS^1(I2)$, $CS^2(I2)$, and $CS^3(I2)$.

It can be observed that with the assistance of the semantics branch, the changed multilevel contextual semantic feature maps can be obtained via 1.2 M parameters.

As we know, the changed spatial details and the changed multilevel contextual semantics exist in the serious semantic gap and resolution gap, so we adopt the DSAM to ensure efficient communication and fusion between the two branches. The DSAM can be seen in Fig. 4, which consists of depthwise convolutions, traditional convolutions, upsample, average pooling, sigmoid function, and batch-normalization operations. It is noted that we just aggregate the changed spatial details C_SD and one of the changed contextual semantics C_CS^3 to obtain changed detail-semantic feature maps C_DSF^1, rather than separately aggregating them at each layer of the changed contextual semantics, as shown in the following equation:

$$C_\mathrm{DSF}^1 = \mathrm{DSAM}([C_SD, C_SC^3]). \tag{1}$$

B. Cross-Scale Coupling Module

Due to the partial expansion and reduction of buildings, along with the construction and demolition of buildings, BCD poses higher demands on the scale coverage capability of the model. Although the semantics branch has generated changed multilevel contextual semantic feature maps C_CS^1, C_CS^2, and C_CS^3, those changed feature maps remain independent and never undergo depth fusion. Therefore, to further enhance the ability to capture multiscale changed buildings, we propose a CSCM to effectively combine those changed feature maps with different scales and semantic levels from bottom to top,

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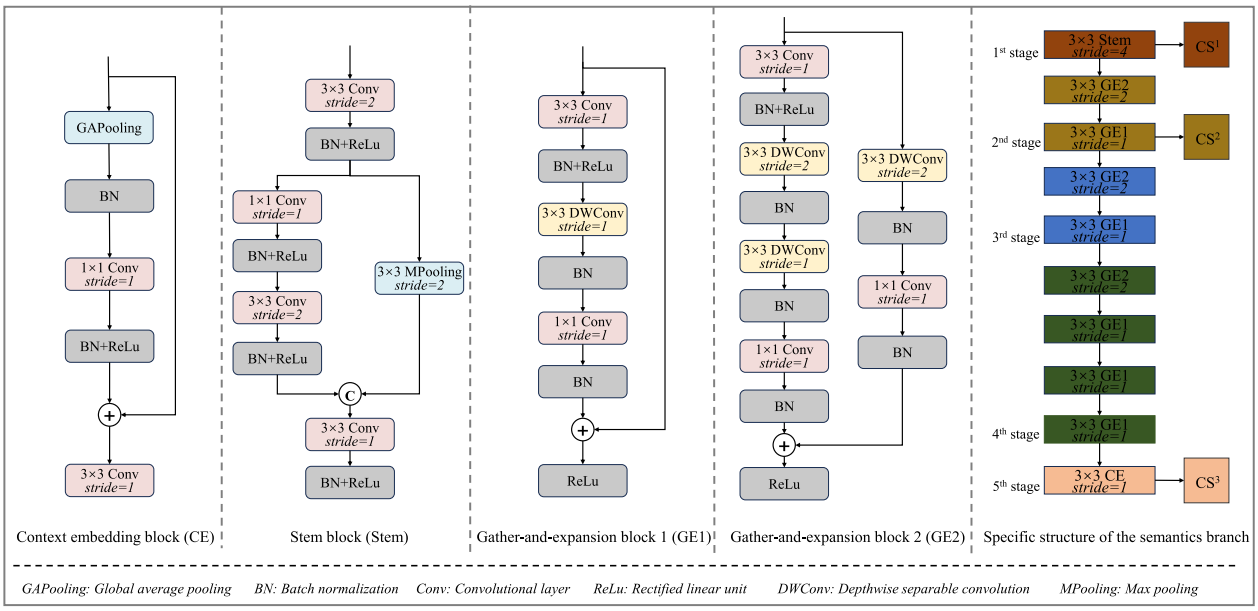


Fig. 3. Components of the semantics branch.

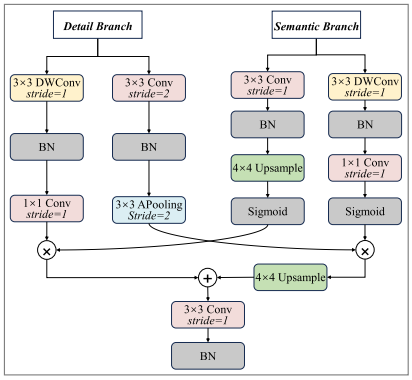


Fig. 4. Detail-semantics aggregation module.

ensuring the richness of multiscale information. Concretely, the CSCM consists of a convolutional layer followed by batch normalization, upsampling operation, and activation function. As in formulas (2) and (3), first, after obtaining the detail-semantics feature maps C_DSF^1 via the DSAM, we fuse it with C_CS^2 to generate the fined changed feature maps C_DSF^2; then, we once fuse C_DSF^2 with C_CS^1 and upsample them to the resolution of original images to obtain the final fine-grained and high-level semantic changed feature maps C_DSF. By following the approach, we not only retain detailed information at each scale feature map but also achieve deep fusion of changed feature maps at different levels

$$C_\mathrm{DSF}^2 = \mathrm{Bn}(\mathrm{ReLu}(F(\mathrm{Cat}[C_\mathrm{DSF}^1, C_CS^2]))) \tag{2}$$

$$C_\mathrm{DSF} = \mathrm{Up}(\mathrm{Bn}(\mathrm{ReLu}(F(\mathrm{Cat}[\mathrm{Up}(C_\mathrm{DSF}^2), C_CS^1])))) \tag{3}$$

where ReLu represents the activation function, Bn represents the batch normalization, F represents the convolution operation, Cat represents the concatenation of two feature maps, and Up represents the upsampling operation.

C. Boundary Constraint Module

As is well known, the boundaries of changed buildings are complex caused by the accompanying shadows and occlusion from other land covers. However, the current existing BCD primarily emphasizes the regional accuracy of changed buildings and often overlooks the boundary accuracy that reflects their geometric and spatial location. In this article, although the CSCM has captured rich multiscale information, recovering refined details at the boundaries still remains challenging due to the multiple pooling and convolution operations. Therefore, we introduce a BCM to improve refined boundary detection. Specifically, the principle of the BCM is as follows. First, we use the ground truth of changed buildings to generate the ground truth of changed building boundaries; then, we use the output of the CSCM to generate the changed maps of boundaries; next, in the training phase, the ground truth of changed building boundaries are used to directly supervise the changed maps of boundaries; and in the testing phase, the changed maps of boundaries are merged with the changed maps of buildings to obtain the final prediction maps of changed buildings.

The procedure for generating the ground truth of changed building boundaries is illustrated in Fig. 5. Based on Laplacian convolutional kernels (as shown for Kernel_1 in Fig. 5), we can obtain boundary maps of changed buildings at different scales by varying the strides. Subsequently, the boundary maps at different scales are upsampled to restore to the original size. Finally, a weighted calculation (as shown for Kernel_2 in Fig. 5) is applied to the boundary maps at different scales to obtain the ground truth of changed building boundaries.

D. Loss Function

The loss function consists of two parts: the cross-scale coupling loss function and the boundary constraint loss function. In the cross-scale coupling loss function, we choose the

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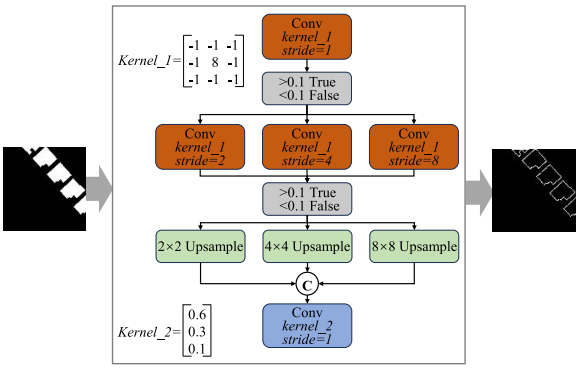


Fig. 5. Procedure for generating the ground truth of changed building boundaries.

cross-entropy loss function as the loss function. It can be expressed as formula (4), where N represents the total number of pixels, $t_i \in \{0, 1\}$ represents the value of the ith pixel (i.e., $t_i = 0$ and $t_i = 1$ represent the nonchanged building and changed building in the ith pixel, respectively), and $p_i \in (0, 1)$ and $(1 - p_i) \in (0, 1)$ represent the predicted probabilities of changed building and nonchanged building of the ith pixel, respectively,

$$L_{s-\text{bce}} = -\frac{1}{N}\sum_{i}^{N}\left[t_i \log(p_i) + (1 - t_i)\log(1 - p_i)\right]. \quad (4)$$

In the boundary constraint loss function, given the significant imbalance in the number of changed and unchanged categories, we use the dice loss function and the cross-entropy loss function to address this issue. It can be expressed as formulas (5) and (6), where $t_i' \in \{0, 1\}$ represents the value of the ith pixel (i.e., $t_i = 0$ and $t_i = 1$ represent the boundaries of nonchanged building and changed building in the ith pixel, respectively) and $p_i' \in (0, 1)$ and $(1 - p_i') \in (0, 1)$ represent the predicted probabilities in the boundary of nonchanged building and changed building of the ith pixel, respectively,

$$L_{\text{dice}} = 1 - \frac{\sum_{i=1}^{N}\left(t_i' * p_i'\right)}{\sum_{i=1}^{N} t_i' + \sum_{i=1}^{N} p_i'} \quad (5)$$

$$L_{b-bce} = -\frac{1}{N}\sum_{i}^{N}\left[t_i' \log\left(p_i'\right) + \left(1 - t_i'\right)\log\left(1 - p_i'\right)\right]. \quad (6)$$

Therefore, the hybrid loss function of the network can be expressed as formula (7), where the weight ratio γ is used to balance the influence of changed building and changed boundary, which is set to 4 (the analysis of the hyperparameters are presented in Section III-D6)

$$L = L_{s-\text{bce}} + \gamma * (L_{\text{dice}} + L_{b-\text{bce}}). \quad (7)$$

III. Experimental Results and Analysis

A. Dataset Description

In this article, we utilize two public BCD datasets as experimental datasets: the WHU dataset [29] and the LEVIR dataset [30], as shown in Fig. 6.

The WHU dataset comprises one pair of aerial and satellite images of size 32 507 × 15 354 with a resolution of 0.075 m, covering Christchurch, New Zealand, from 2012 to 2016. Among them, we select 5334, 762, and 1524 image pairs as training, validation, and testing, respectively. The dataset features various sizes and types of changed buildings, including circular tank buildings, rectangular warehouses, and irregular residences with roofs of different colors. In addition, it includes numerous irrelevant changed objects, such as parks, roads, and cars.

The LEVIR dataset comprises 637 very high-resolution Google Earth image pairs of size 1024 × 1024 with a resolution of 0.5 m, covering several cities in Texas of USA from 2002 to 2018. Also, we cut both original image pairs into nonoverlapping tiles of 256 × 256 pixels. Among them, we select 7120, 1024, and 2048 image pairs as training, validation, and testing, respectively. The dataset is characterized by blurry boundaries and the small size of villa residences and apartments, stemming from the low spatial resolution of images, serious shadows, and the dense distribution of buildings.

B. Evaluation Metrics and Parameter Setting

For an objective evaluation of the BCD results, the precision (Pre), recall (Rec), $F1$ score ($F1$), and intersection over union (IoU) are selected as the assessment metrics. TP, FP, and FN represent the true positive, false positive, and false negative, respectively,

$$\text{Precision} = \frac{\text{TP}}{\text{TP} + \text{FP}} \quad (8)$$

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}} \quad (9)$$

$$F1 = 2 * \frac{\text{Precision} * \text{Recall}}{\text{Precision} + \text{Recall}} \quad (10)$$

$$\text{IoU} = \frac{\text{TP}}{(\text{TP} + \text{FN} + \text{FP})}. \quad (11)$$

Our proposed method is implemented with the following parameter settings: the SGD optimization algorithm with a momentum of 0.9, a weight decay of $5e^{-4}$, and a learning rate of 0.0001. During the training stage, we run 200 epochs with a batch size of 4. For the comparative methods, we follow the parameter settings outlined in the original paper. Furthermore, all experiments are conducted on the Pytorch platform using an NVIDIA RTX3080 GPU with 12 GB of memory.

C. Comparison Experiments

In this section, we select 14 state-of-the-art changed detection networks as comparison methods to evaluate our proposed method from both accuracy and efficiency (the detailed description of comparison methods is shown in Table I). These comparison methods can be broadly categorized into seven nonlightweight networks and seven lightweight networks based on whether the number of parameters exceeds 10 M. Also, for a more detailed comparison with our proposed detail branch, semantics branch, CSCM, and BCM, these methods can be further divided into multiscale modules [e.g., methods of (1), (2), and (10)], boundary refinement modules [e.g.,

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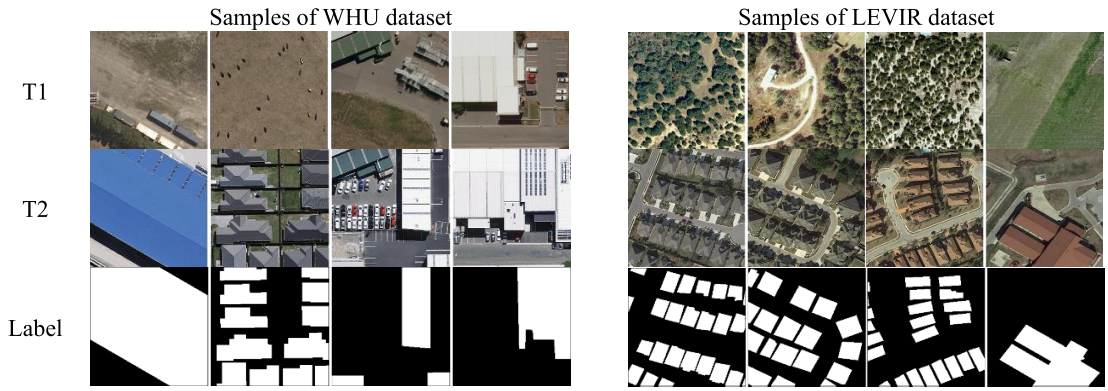


Fig. 6. Sample of WHU and LEVIR datasets.

TABLE I
COMPARISON METHODS OF NON LIGHTWEIGHT AND LIGHTWEIGHT

Methods	Lightweight	Network Description
(1) STANET [30]	×	It is a dense encoder-decoder network architecture for change detection, which integrates a multi-scale module along with a self-attention mechanism.
(2) SNUNET [31]	×	It is a dense encoder-decoder network architecture designed for change detection, which aggregates multi-level semantic feature maps.
(3) IFN [32]	×	It is a deep encoder-decoder network architecture for change detection. The attention module isolates crucial features while employing a deep supervision network to enhance the network's capacity for discrepant information.
(4) CHANGE_EX [24]	×	It employs a series of alternative interaction layers to extract features and an interactive fusion module to allow interactive alignment and feature fusion.
(5) WNET [33]	×	It integrates CNNs and Transformers to extract multilevel local fine-grained features and global long-range contextual dependencies. Additionally, it incorporates a difference enhancement module and a CNN-Transformer fusion module to generate and merge the difference feature maps at various levels, respectively.
(6) TFIGRNET [34]	×	It designs a temporal feature interaction module to enhance interaction between bi-temporal features and capture temporal difference information at diverse feature levels. Furthermore, a guided refinement module is used to aggregate both low- and high-level temporal difference representations to polish the location information of high-level features and filter the background clutters of low-level features.
(7) TCDNET [13]	×	It proposes a multi-frequency channel attention module to boost the ability of modeling the channel correlation. Additionally, it presents a joint multi-frequency difference feature enhancement guiding block to improve the boundary quality and the positioning awareness of truly changed objects, respectively.
(8) FC_EF [35]	√	It is a lightweight network encoder-decoder architecture for change detection, where it stacks bi-temporal images as its input to carry out change detection tasks.
(9) BIT [36]	√	It is a lightweight network architecture built on the Transformer framework, which converts input images into a small number of tokens and models the semantics within a compact token-based spacetime for change detection.
(10) LIGHTNET [22]	√	It utilizes MobileNetV2 as its backbone, integrating multi-level semantic features with multi-scale contextual information to generate high-resolution semantic changed feature maps.
(11) FFBDNET [23]	√	It leverages EfficientNet as its foundation, introducing a feature interleaved fusion module to merge shared feature information with difference variance feature information.
(12) DMINET [37]	√	It proposes an intertemporal joint-attention block to steer the global feature distribution of each input. In addition, a reliable architecture is designed by spotlighting the difference acquisition using subtraction and concatenation as well as the multilevel difference aggregation using incremental feature alignment.
(13) EATDNET [38]	√	It consists of a Siamese encoder and an edge-aware decoder. The former aims to capture the local and global information and mines the temporal clues among bi-temporal RS images, and the latter is used to enhance the refined features of edges.
(14) VCTNET [25]	√	Each pixel of the feature map is treated as a graph node, and a graph neural network is introduced to model structured information for coarse change map prediction. Subsequently, reliable tokens are strengthened by employing self/cross attention schemes, followed by interaction with original features through an anchor-primary attention learning module.

methods of (7) and (13)], dual-branch architecture [e.g., methods of (5)], multilevel information fusion [e.g., methods of (5), (6), and (11)], and transformer architecture [e.g., methods of (5), (9), and (14)].

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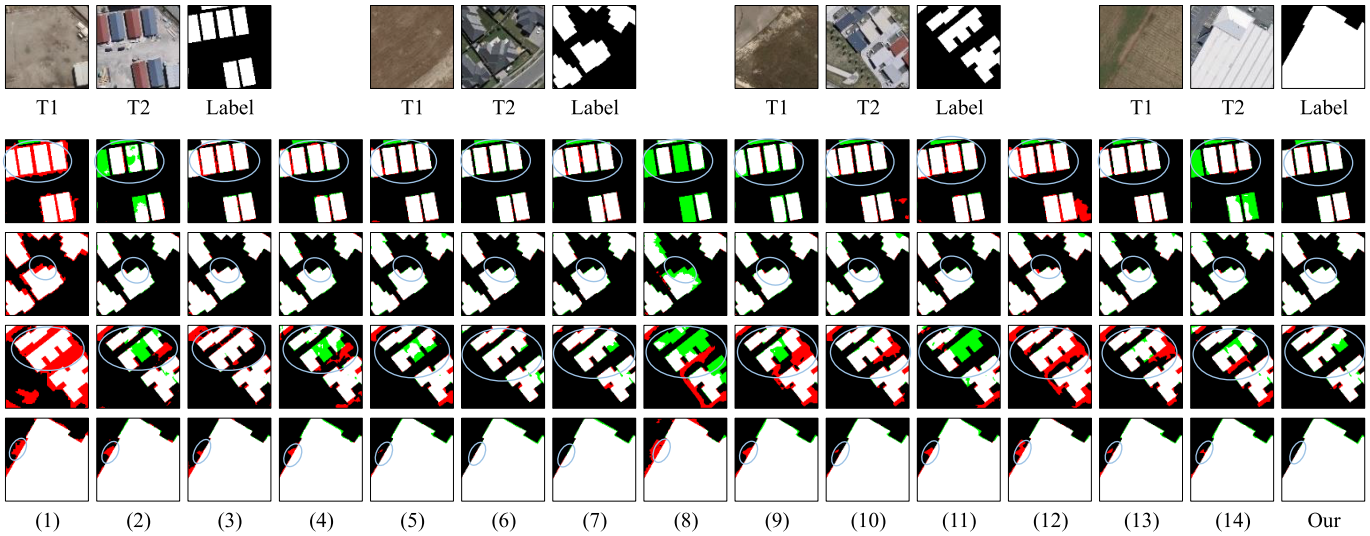


Fig. 7. BCD results of different methods on the WHU datasets. (Red pixels represent false detection areas, green pixels represent missed detection areas, white pixels represent correctly detected areas, and black pixels represent background areas. Subsequent figures adhere to the same color convention.)

In addition, from the accuracy perspective, we evaluate the performance of various methods according to quantitative and qualitative results; from the efficiency perspective, we utilize the number of parameters (Params) and floating-point operations (FLOPs) in the models as criteria to conduct efficiency comparison experiments.

1) Quantitative Evaluation on the WHU Dataset: According to the results of WHU dataset in Table II, our method outperforms both nonlightweight and lightweight networks in terms of *F1* and IoU, achieving remarkable results of 94.41% and 89.41%, respectively. Also, the Params and FLOPs of our method are maintained at a very low level (the detailed description of the efficiency comparison will be provided in Section III-C5). Although our method has not attained the highest precision and recall, the deviation from the best performance is negligible. These indicate that our method excels in accuracy. Furthermore, we observe that nonlightweight networks, such as WNET, TFIGRNET, and TCDNET, achieve good results, with IoU generally exceeding 84%, while lightweight networks, such as LIGHTNET, FFBDNET, EATDNET, and VCTNET, also perform well, with IoU generally exceeding 83%. Notably, except for the TFIGRNET, our approach achieves a substantial improvement of at least 3.8% in IoU compared to other methods.

2) Quantitative Evaluation on the LEVIR Dataset: According to the results of LEVIR dataset in Table II, compared to nonlightweight networks, although our method does not reach the best results in terms of $F1$ and IoU, it achieves the second-best performance. Also, the Params and FLOPs of our method are maintained at a very low level (the detailed description of the efficiency comparison will be provided in Section III-C5). In comparison with lightweight networks, our method exhibits the best performance in both $F1$ and IoU, reaching 91.21% and 83.85%, respectively. In addition, we observe that in terms of IoU, except for the TCDNET method among nonlightweight methods, our method achieves progress of at least 0.2% compared to other nonlightweight methods. Furthermore, compared to lightweight networks, except for the EATDNET method, our method achieves significant progress of about 1%. These findings underscore that our method is positive in accuracy.

In conclusion, combining the quantitative results from the two datasets, the following observations can be made.

1) In the WHU dataset, our method achieves the best performance in both lightweight and nonlightweight networks. However, in the LEVIR dataset, although our method does not achieve the best performance in the nonlightweight network, it still maintains the second-best results, and it continues to excel in the lightweight network. Through in-depth analysis, we find that the phenomenon mainly arises from two aspects. On the one hand, the low resolution of LEVIR dataset and the dense distribution of buildings lead to blurred boundaries of changed buildings and serious shadow effects between buildings, making it difficult for the proposed BCM to effectively fulfill its role. On the other hand, the buildings in the LEVIR dataset are generally small in size and do not form a full-scale distribution, which hinders the adequate performance of the proposed multiscale coupling module.
2) Different state-of-the-art methods exhibit varying performances on different datasets. For example, the TCDNET, which performs best on the LEVIR dataset, does not exhibit equivalent performance on the WHU dataset. Similarly, the VCTNET, which performs well on the WHU dataset, does not demonstrate the same level of performance on the LEVIR dataset. However, both our proposed method and the TFIGRNET show good performance on different datasets. This phenomenon suggests that our proposed lightweight dual branch demonstrates strong robustness, which can extract effective and fine-grained changed feature maps in different datasets, demonstrating robustness.

3) Qualitative Evaluation on the WHU Dataset: We selected four typical image pairs as examples (as shown in the first row of Fig. 7). From the predicted maps, it can be

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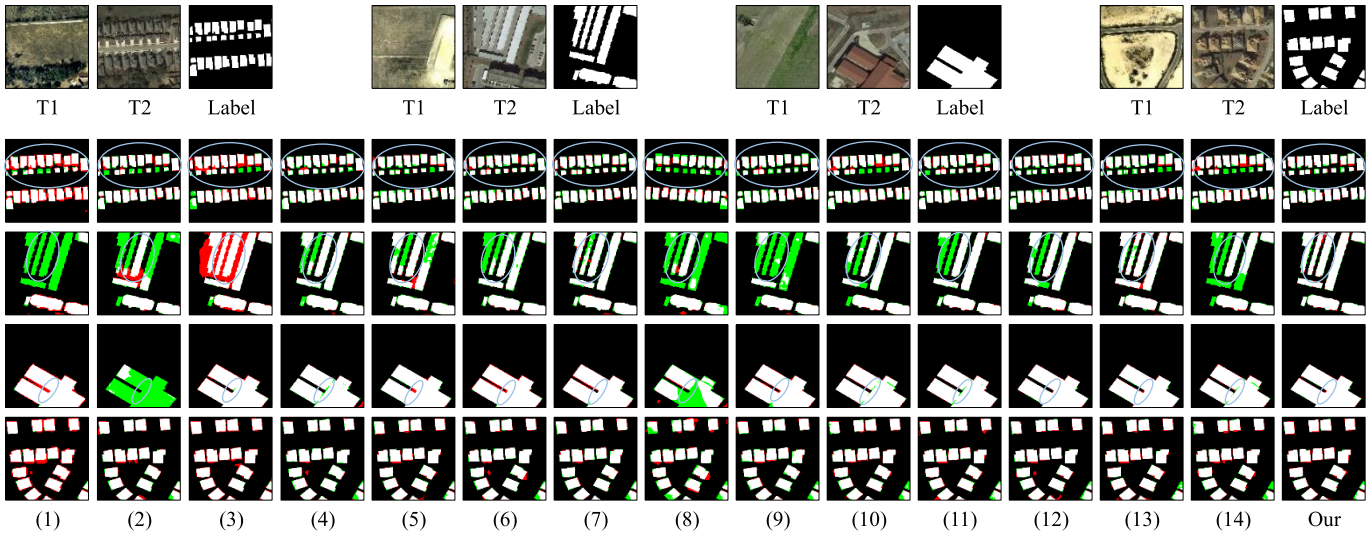


Fig. 8. BCD results of different methods on the LEVIR datasets.

observed that our method and TRIGRNET produce the most fine-grained maps of changed buildings. The outcomes exhibit highly detailed representations of changed buildings at various scales and with significant spectral differences. Moreover, the predicted maps align with the conclusions drawn from our quantitative results.

In detail, we can find that the following conditions hold.

1) As shown in the second row (characterized by significant spectral differences in changed buildings, such as red roofs and gray-brown roofs), the results demonstrate that except for STANET, SNUNET, FC_EF, DMINET, and VCTNET, other methods can recognize changed buildings with significant spectral differences. Also, our method and TRIGRNET show superior performance, excelling both in the boundaries and the regions of changed buildings.
2) As shown in the third row (characterized by complex boundaries of changed buildings), the results reveal that FC_EF and STANET can only approximate the outlines of the changed buildings and fail to capture the boundaries of changed buildings. However, among the remaining methods, TFIGRNET and our method particularly outperform in accurately detecting changed buildings and closely approximating the ground truth map of changed buildings.
3) As shown in the fourth row (characterized by significant spectral differences in changed buildings, as well as road interference and complex boundaries), the results demonstrate that each method has serious false detections and missed detections. In comparison, our method, TFIGRNET, and TCDNET achieve the best performance, with only a few areas of false identification.
4) As shown in the fifth row (characterized by large-sized changed buildings), our method, TFIGRNET, and TCDNET perform well in handling large changed buildings. It not only detects the completely changed buildings but also effectively avoids interference of background from the surrounding of changed buildings.

4) Qualitative Evaluation on the LEVIR Dataset: We selected four typical image pairs as examples (as shown in the first row of Fig. 8). From the predicted maps, it can be observed that our method obtains the best performance. Despite a slight gap in visual results compared to the WHU dataset, the overall performance is still acceptable. Moreover, our method exhibits satisfactory results in handling changed buildings of various scales, different roof types, and significant shape differences.

In detail, we can find that the following conditions hold.

1) As shown in the second row (characterized by small-scale changed buildings), the results demonstrate that except for TFIGRNET, TCDNET, DMINET, and our method, other methods generally exhibit severe missed detections on small-scale changed buildings. On the other hand, it can also be seen that in the scenarios with densely distributed buildings, TCDNET and DMINET show some slight false detections compared to our method.
2) As shown in the third row (characterized by changed buildings with elongated shapes, as well as significant spectral differences with white, gray, and dark colors), it can be found that except for EATDNET and our method, other methods suffer from severe false detections. It indicates that for extremely complex cases of changed buildings, the detection ability of the other methods still needs further improvement.
3) As shown in the fourth row, most state-of-the-art methods have a common issue of identifying the shadows over the roofs of changed buildings as nonchanged buildings. However, our method can effectively avoid this situation and obtain a complete and clear changed map.
4) As shown in the fifth row, for some changed buildings that are similar to the background (such as brownish-yellow buildings resembling the ground), our method achieves good results without significant areas of misidentification.

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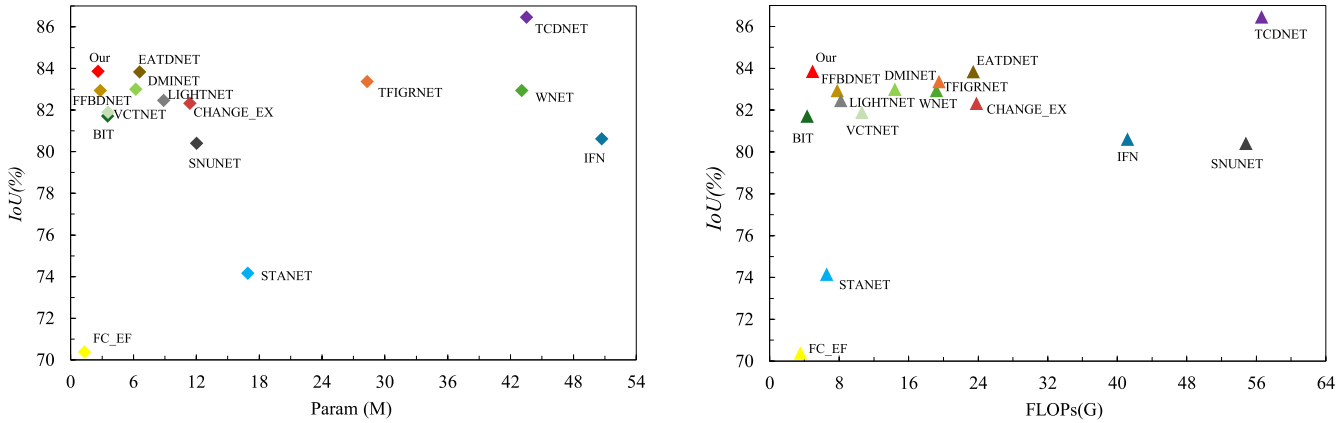


Fig. 9. Relationships between Params and IoU, as well as between FLOPs and IoU.

Therefore, we can draw the following conclusion from the visual results of the two datasets.

1) FC_EF, SNUNET, and STANET exhibit a comparatively weaker generalization ability in changed buildings. Through analysis, we can know that the main reason is that those networks fail to effectively capture the high-resolution changed feature maps.
2) BIT, CHANGE_EX, LIGHTNET, and FFBDNET are capable of recognizing changed buildings with various sizes, to some extent. However, for the extremely complex changed building scenarios, these networks exhibit insufficient stability to generate satisfactory outcomes (such as the visual results of the fourth row in the WHU and the third row in the LEVIR visual results). Through in-depth analysis, we can know that the main for this phenomenon is that those networks fail to capture sufficiently fine-grained and high-level semantic changed feature maps, especially at the boundaries of changed buildings, despite the inclusion of auxiliary modules such as multiscale and multilevel feature fusion modules.
3) The nonlightweight TCDNET method demonstrates exceptional visualization results. Through the analysis, we can know that the main for this phenomenon is that the multifrequency channel attention module can boost the ability to model the channel correlation. In addition, the joint multifrequency difference feature enhancement guiding block can improve the boundary quality and the positioning awareness of truly changed objects.
4) In addition, we observe that our method outperforms other methods in capturing the boundaries of changed buildings at the two datasets. The main reasons originate from two aspects. On the one hand, the proposed detail branch can maintain the rich spatial details of changed buildings. On the other hand, the proposed BCM plays a positive role and improves the sensitivity to the boundaries of changed buildings.
5) Moreover, we observe that our method can effectively identify changed buildings with various sizes, complicated shapes, and diverse colors of roofs in the two datasets. The main reasons originate from two aspects. On the one hand, the proposed semantics branch can extract high-level semantic feature maps of changed buildings. On the other hand, the CSCM further enriches semantic and scale representation ability from the scale perspective.
6) However, it cannot be ignored that our method does not perform perfectly in the extremely complex examples, such as the fourth row in the WHU result, which indicates that there is still room for further improvement for our method.

5) Efficiency Test: To comprehensively assess the complexity of the model, we select the number of parameters and FLOPs as evaluation criteria to compare the efficiency of various methods. The number of parameters is used to represent the scale of parameters in the model, while the FLOPs are used to measure the overall computational workload in the model. Table III displays the IoU, Params, and FLOPs for each method. To provide a more intuitive representation, we also plotted the relationships between Params and IoU, as well as FLOPs and IoU, as shown in Fig. 9.

From the table and figure, the following observations can be made.

1) FC_EF has the smallest number of Params and FLOPs, with 1.35 M and 3.58 G, respectively. However, its accuracy is the lowest, potentially making it challenging to meet practical requirements.
2) TCDNET achieves the highest IoU, reaching 86.45%. Nevertheless, its FLOPs and Params locate at the highest level, significantly surpassing other lightweight networks.
3) Next, in the nonlightweight network, CHANGE_EX, WNET, and TFIGRNET achieve similar levels of accuracy, ranging between 82.3% and 82.93%, with their Params and FLOPs falling within a comparable range.
4) Then, in the lightweight network, VCTNET, FFBDNET, LIGHTNET, and BIT exhibit similar levels of Params and FLOPs ranging between 2.8–9 M and 4.3–11.7 G, with their accuracy falling within a comparable range.
5) Except for TCDNET and FC_EF, our method achieves the highest IoU and the smallest Params and FLOPs, with 2.64 M, 4.95 G, and 83.85%, respectively.

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TABLE II

BCD ACCURACY OF DIFFERENT METHODS ON THE WHU AND LEVIR DATASETS (RED FONTS REPRESENT OPTIMAL PERFORMANCE AND BLUE FONTS REPRESENT SUBOPTIMAL PERFORMANCE. SUBSEQUENT TABLES ADHERE TO THE SAME COLOR CONVENTION)

		WHU				*LEVIR*					
	Methods	*Pre*	*Rec*	*F1*	*IoU*	*Pre*	*Rec*	*F1*	*IoU*	*Params (M)*	*FLOPs (G)*
Non-Lightweight Networks	(1) STANET [30]	66.32	**94.11**	77.81	63.68	77.98	**93.78**	85.16	74.15	16.93	6.58
	(2) SNUNET [31]	90.24	90.32	90.28	82.28	89.17	89.10	89.13	80.40	12.03	54.83
	(3) IFN [32]	89.51	91.53	90.51	82.66	86.22	**92.54**	89.27	80.61	50.71	41.18
	(4) CHANGE_EX [24]	94.51	88.10	91.20	83.82	92.48	88.22	90.30	82.32	11.39	23.82
	(5) WNET [33]	93.52	89.42	91.43	84.21	91.16	90.18	90.67	82.93	43.07	19.20
	(6) TFIGRNET [34]	**95.86**	92.68	**94.25**	**89.12**	92.15	89.73	90.93	83.36	28.32	19.52
	(7) TCDNET [13]	**96.55**	88.31	92.25	85.61	**93.55**	91.54	**92.53**	**86.45**	43.54	56.64
Lightweight Networks	(8) FC_EF [35]	81.16	71.00	75.74	60.96	87.00	78.65	82.62	70.38	**1.35**	**3.58**
	(9) BIT [36]	92.80	88.81	90.76	83.09	91.91	88.04	89.93	81.71	3.55	4.35
	(10) LIGHTNET [22]	94.37	90.12	92.20	85.53	92.12	88.71	90.39	82.46	8.9	8.21
	(11) FFBDNET [23]	92.12	89.95	91.02	83.52	92.21	89.18	90.67	82.93	2.85	7.81
	(12) DMINET [37]	86.86	91.96	89.34	80.73	92.52	89.95	90.71	82.99	6.24	14.42
	(13) EATDNET [38]	94.22	87.61	90.80	83.14	91.74	90.67	91.20	83.83	6.59	23.43
	(14) VCTNET [25]	94.42	88.93	91.59	84.48	**92.57**	87.65	90.04	81.89	3.57	10.64
	Our method	95.71	**93.14**	**94.41**	**89.41**	92.30	90.15	**91.21**	**83.85**	**2.64**	**4.95**

6) Finally, although the Params and FLOPs of our method do not reach the minimum values, the differences of Params and FLOPs compared to the minimum values are relatively small, and our accuracy still remains at the highest level.

In summary, while the Params and FLOPs of our method do not reach the minimum values, the differences compared to the minimum values are relatively small. In addition, despite a 2.6% disparity in IoU compared to TCDNET, our method achieves significantly lower Params and FLOPs. These efficiency comparison experiments clearly demonstrate that our method maintains exceptional efficiency without sacrificing detection accuracy, affirming the lightweight nature of our proposed network.

D. Ablation Experiments

To comprehensively evaluate and analyze the effectiveness of our proposed method, we conduct ablation experiments on the LEVIR dataset from both quantitative and qualitative perspectives, including the effectiveness of different branches, the effectiveness of the CSCM, the effectiveness of the BCM, as well as the effectiveness of the combining of modules on dual branch. Moreover, to substantiate the rationale behind integrating multilevel semantic feature maps and selecting the weight parameter γ in the loss function, we also conduct corresponding ablation experiments.

1) Effectiveness of Different Branches: From the quantitative results of indexes 1, 4, and 7, it can be found that relying solely on the detail branch or semantics branch proves insufficient for achieving satisfactory results, with the IoU reaching only 68.14% and 76.98%, respectively. However, the dual-branch network achieves a higher IoU of 80.95%, at least 5% higher than the formers. On the other hand, the recall on the detail branch and the precision on the semantics branch are very low, only 76.43% and 83.45%, respectively, which are significantly lower compared with the recall and precision of the dual-branch network, respectively.

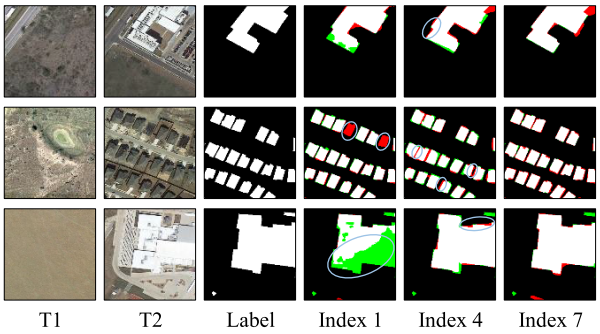


Fig. 10. Result of different branches.

Furthermore, it can be observed from the visual results in Fig. 10. In the detail branch, significant issues of false detections and missed detections have occurred. Specifically, two small-scale nonchanged buildings have been mistakenly identified as changed buildings, and some areas of the large-scale changed buildings have been mistakenly identified as nonchanged buildings (as shown in the blue circles of the detail branch). Conversely, in the semantics branch, false alarms and missed alarms have been mitigated slightly. For instance, two small-scale nonchanged buildings are filtered out, and the large-scale changed buildings are effectively extracted in their entirety. However, it cannot be ignored that the boundaries of the changed buildings in the semantics branch appear excessively blurry, accompanied by instances of sticking together (as shown in the blue circles of the semantics branch). Finally, the combination of the two branches has yielded favorable visual results. Not only have the occurrences of false alarms and missed alarms been substantially reduced, but the boundaries of changed buildings have also become more refined.

Therefore, we can draw the following conclusions. First, relying solely on the detail branch or the semantics branch

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TABLE III
MODEL COMPLEXITY COMPARISONS ON THE LEVIR DATASET

Methods	*IoU*	*Params(M)*	*FLOPs(G)*
(1) STANET [30]	74.15	16.93	6.58
(2) SNUNET [31]	80.40	12.03	54.83
(3) IFN [32]	80.61	50.71	41.18
(4) CHANGE_EX [24]	82.32	11.39	23.82
(5) WNET [33]	82.93	43.07	19.20
(6) TFIGRNET [34]	83.36	28.32	19.52
(7) TCDNET [13]	**86.45**	43.54	56.64
(8) FC_EF [35]	70.38	**1.35**	**3.58**
(9) BIT [36]	81.71	3.55	4.35
(10) LIGHTNET [22]	82.46	8.9	8.21
(11) FFBDNET [23]	82.93	2.85	7.81
(12) DMINET [37]	82.99	6.24	14.42
(13) EATDNET [38]	83.83	6.59	23.43
(14) VCTNET [25]	81.89	3.57	10.64
Our method	**83.85**	**2.64**	**4.95**

TABLE IV
ACCURACY OF ABLATION EXPERIMENTS ON DIFFERENT BRANCHES, CSCM, AND BCM

Index	*Detail branch*	*Semantics branch*	*CSCM*	*BCM*	*Pre*	*Rec*	*IoU*
1	√	×	×	×	86.26	76.43	68.14
2	√	×	√	×	89.23	83.70	76.02
3	√	×	×	√	87.16	80.28	71.79
4	×	√	×	×	83.45	90.85	76.98
5	×	√	√	×	87.63	89.34	79.33
6	×	√	×	√	90.96	87.17	80.22
7	√	√	×	×	88.61	90.23	80.95
8	√	√	√	×	91.20	89.09	82.03
9	√	√	×	√	89.24	**91.33**	82.27
10	√	√	√	√	**92.30**	90.15	**83.85**

TABLE V
ACCURACY OF INTEGRATING MULTILEVEL SEMANTIC FEATURE MAPS

Index	*1st*	*2nd*	*3rd*	*4th*	*IoU*	*Params(M)*	*FLOPs(G)*
1	√	√	√	√	83.47	4.07	6.43
2	×	√	√	√	83.07	3.19	5.52
3	√	×	√	√	82.97	3.52	5.86
4	√	√	×	√	83.13	3.90	6.70
5	√	√	√	×	83.22	4.03	6.24
6	×	×	√	√	82.20	3.78	6.21
7	√	√	×	×	**83.85**	**2.64**	**4.95**

struggles to achieve satisfactory results in BCD research. Then, the high-level contextual semantics provided by the semantics branch play a more positive role, compared with the low-level spatial details provided by the detail branch. Finally, the dual branch is feasible for BCD research, as it effectively integrates the advantages of the two branches.

2) Effectiveness of CSCM: In the evaluation of the CSCM, we conduct experiments separately on the detail branch, detail branch with CSCM, semantics branch, semantics branch with CSCM, dual branch, and dual branch with CSCM. The quantitative results are shown in Table IV of indexes 1, 2, 4, 5, 7, and 8. It can be found that, with the help of the CSCM, the IoU generally shows significant improvement. Concretely, the precision has improved with increments of 3%, 4.2%, and 2.6% in the detail branch, semantics branch, and dual branch, respectively. Nevertheless, the recall does not exhibit a consistent improvement and even experiences a slight decrease.

Furthermore, it can be observed from the visual results in Fig. 11. First, compared with the detail branch, despite the existence of some false alarms and missed alarms, the overall coherence of the large-scale changed building in the detail branch with CSCM is maintained and a small-scale unchanged building is successfully filtered out (as shown in the blue circles of the detail branch with CSCM). Second, compared with the semantics branch, while there is also an increase in false alarms in the semantics branch with CSCM (e.g., the yellow circles, incorrectly identifying the area as changed buildings), the boundaries of changed buildings become more refined (e.g., the blue circles, avoiding the instances of changed buildings sticking together). Finally, compared with the dual branch, the dual branch with CSCM contributes to significant improvements in false alarms and missed alarms. As shown in the blue circles of the dual branch with CSCM, the missing changed building has been accurately identified. Also, the boundaries of changed buildings are more refined, and false alarms have decreased.

Therefore, we can draw the following conclusions. On the one hand, integrating the CSCM into the detail branch effectively enhances the identification ability of the network to changed buildings at various scales. However, adding the CSCM into the semantics branch may marginally enhance the boundary quality but does not comprehensively improve the overall performance of BCD. On the other hand, integrating the CSCM into the dual branch enhances the sensitivity of the network to small-scale changed buildings, aligning with the demand for multiscale characteristics in changed buildings.

3) Effectiveness of BCM: Similarly, in the evaluation of the BCM, we conduct experiments separately on the detail branch, detail branch with BCM, semantics branch, semantics branch with BCM, dual branch, and dual branch with BCM. The quantitative results are shown in Table IV of indexes 1, 3, 4, 6, 7, and 9. From the quantitative results, it can be found that, with the help of the BCM, the IoU generally shows significant improvement, with an increase of approximately 2%. In addition, the BCM generally improves both precision and recall in the detail branch with BCM and the dual branch with BCM. However, in the semantics branch with BCM, only the precision has been improved.

Furthermore, it can be observed from the visual results in Fig. 12. First, compared with the detail branch, the visual results have shown a slight improvement in the detail branch with BCM. Second, compared with the semantics branch, the boundaries of changed buildings in the semantics branch with BCM become refined, and the false alarm phenomena at the boundaries have been significantly suppressed (as shown in the blue circles of the semantics branch with BCM). However, it should not be overlooked that a recurrence of the hollow phenomenon within the interior of changed buildings persists (as shown in the yellow circles of the semantics branch with BCM). Finally, compared with the dual branch, the majority

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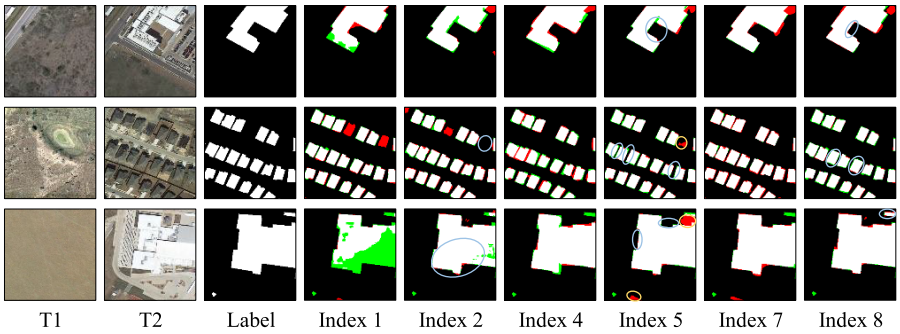


Fig. 11. Result of different branches w or w/o CSCM.

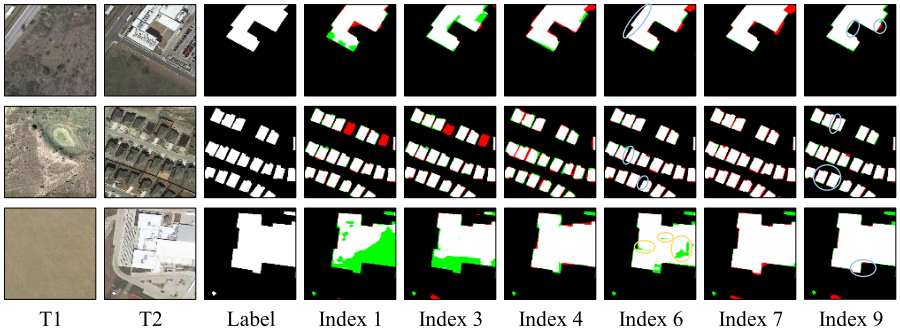


Fig. 12. Result of different branches w or w/o BCM.

of false alarm and missed alarm regions are effectively suppressed, and the boundaries of changed buildings become more refined in the dual branch with BCM (as shown in the blue circles of the dual branch with BCM).

Therefore, we can draw the following conclusions. First, in the detail branch, the addition of the BCM does not effectively enhance the quality of the boundary and the overall performance in BCD. Second, the proposed BCM plays a positive role in the semantics branch, to some extent compensating for deficiencies in detailed information, but there are still instances of instability. Finally, incorporating the BCM into the dual-branch network can effectively refine the boundaries of changed buildings.

4) Effectiveness of the Combining of Dual Branch and Modules: From the quantitative results as shown in Table IV of indexes 7, 8, 9, and 10, it can be found that with the help of the BCM and the multiscale coupling module, our proposed method achieves an increase of approximately 3% in IoU. Moreover, the dual branch with CSCM and the dual branch with BCM achieve similar performance, but the latter has a 2% higher precision and a 2% lower recall compared to the former.

Furthermore, it can be observed from the visual results in Fig. 13. Compared with the dual branch, the dual branch with CSCM has improved both missed alarms and false alarms (as shown in the blue circles of the dual branch with CSCM), the dual branch with BCM has addressed some instances of false alarms (as shown in the blue circles of the dual branch with BCM). Moreover, our method achieves the best visual results with almost no noticeable issues of false alarms and missed alarms (as shown in the blue circles of our method).

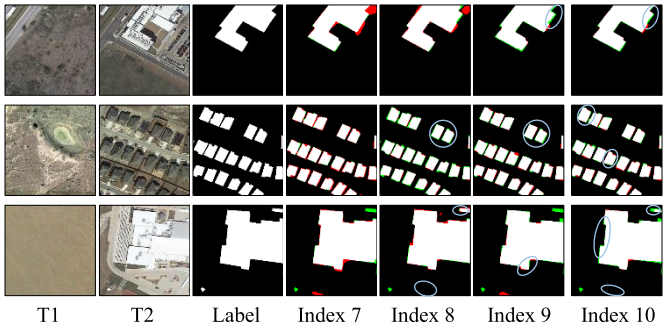


Fig. 13. Result of the combining of dual branch and modules.

The above phenomena illustrate that in the dual-branch structure, the CSCM and the BCM can complement each other effectively, combining their strengths to compensate for the shortcomings of the original dual-branch network in terms of scale and boundaries.

5) Effectiveness of Different Level Semantic Feature Maps: To validate the rationale behind integrating multilevel semantic feature maps, we perform an ablation experiment on the LEVIR dataset. The experimental results are presented in Table V. Here, first, second, third, and fourth correspond to the feature map outputs in Fig. 3, with $\surd$ indicating the inclusion of the feature maps and $\times$ indicating exclusion. The fifth feature map is not considered, as it has already been used in generating the detail-semantic changed feature maps. The experimental results demonstrate that the following conditions hold.

1) Combining the first and second feature maps with the detail-semantic feature maps (our proposed method)

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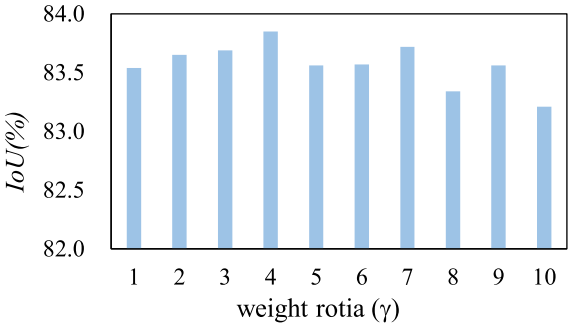


Fig. 14. Performance of different weight parameters.

achieves the highest IoU accuracy and the lowest Params and FLOPs.

2) Combining the first, second, third, and fourth feature maps with the detail-semantic feature maps does not achieve the expected optimal accuracy.
3) Combining the third and fourth feature maps with the detail-semantic feature maps results in less favorable outcomes.

Therefore, in this article, combining the first and second feature maps with the detail-semantic feature map proves effective, achieving the highest IoU while maintaining the lowest Params and FLOPs.

6) Effectiveness of Weight Parameter in the Loss Function: To assess the impact of weight parameters γ, we conduct ablation experiments about the selection of weight parameters in the loss function. Specifically, we set the weight for the boundary constraint loss function to [1], [2], [3], [4], [5], [6], [7], [8], [9], and [10] and analyze their impact on the performances.

The results, as shown in Fig. 14, indicate that different weights lead to varying performances. Despite minor differences, we observed optimal accuracy when the weight was set to 4. Therefore, setting the weight to 4 is a reasonable and effective choice in our model.

IV. Discussion

In this article, we propose a lightweight dual-branch network architecture that integrates the CSCM and the BCM for BCD. Compared with state-of-the-art lightweight and nonlightweight models, our method not only achieves remarkable accuracy but also excels in efficiency. Specifically, we will conduct a comprehensive analysis and discussion from two perspectives.

On the one hand, from the efficiency perspective, unlike other lightweight networks that utilize a single-branch network architecture, we propose a lightweight dual-branch network architecture to address the challenges related to high computation costs and large model sizes. Built upon the network architecture, the spatial details and contextual semantics can be generated rapidly and efficiently. In the detail branch, we can quickly obtain high-resolution feature maps using only three convolutional stages, with a parameter count of just 0.5 M; in the semantics branch, we can efficiently extract high-level deep semantic information by combining the stem block, gather-and-expansion block, and context embedding block, with a parameter count of just 1.2 M. At the same time, the CSCM can further enrich semantic and scale information without significantly increasing the parameters, and the BCM can refine the boundary of changed buildings without compromising the inference speed.

On the other hand, from the accuracy perspective, we observe that even without the involvement of the two modules, the lightweight dual-branch network can also achieve performance comparable to some state-of-the-art lightweight methods, demonstrating that the proposed lightweight dual-branch network architecture exhibits notable superiority. In addition, with the help of the CSCM and BCM, the representation ability across multiscales and the sensitivity to the boundaries of changed buildings in the model can be further improved.

However, it is worth noting that our method did not achieve optimal efficiency and accuracy. Consequently, future research efforts may focus on enhancing both the computational efficiency and stability of the dual-branch network.

V. Conclusion

In this article, we propose a lightweight dual-branch network architecture that integrates both the CSCM and the BCM for BCD. Through utilizing the dual-branch architecture, the challenges related to high computation costs and large model sizes can be effectively mitigated, while the simultaneous capture of spatial details and contextual semantics also becomes feasible. Moreover, to adapt to the multiscale characteristics of changed buildings, the CSCM is introduced to further enrich the semantic and scale representation ability of changed feature maps from a scale perspective. In addition, to alleviate the loss of edge information caused by convolution and pooling operations, the BCM is integrated to further improve the model's sensitivity to the boundaries of changed buildings. Extensive comparative experiments with state-of-the-art lightweight and nonlightweight models have demonstrated that our method not only achieves remarkable accuracy but also excels in efficiency.

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